

Fig 1. Monthly Live Births in South Korea (1997–2024)

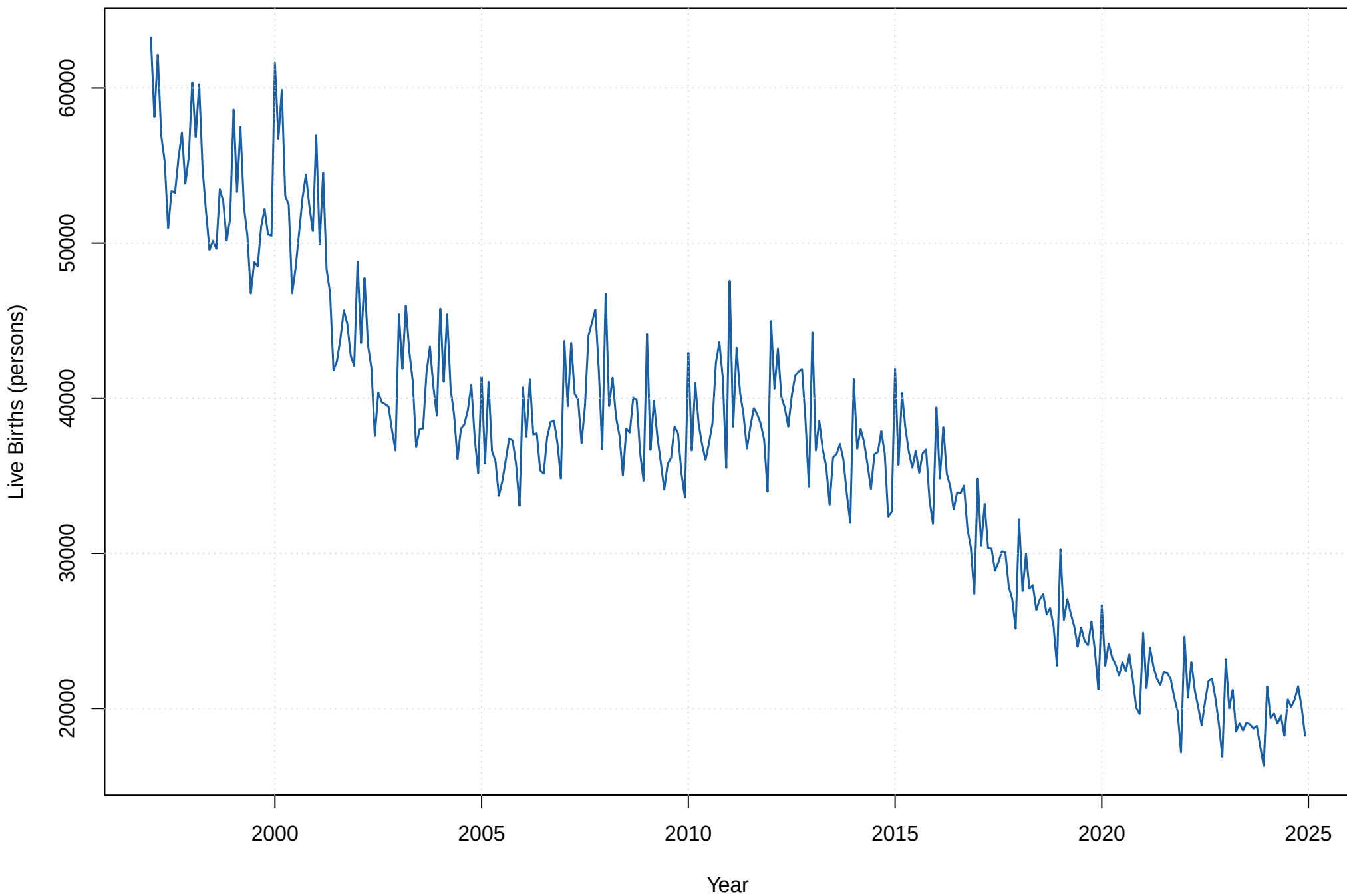


Fig 2. Original Series with 2×12 Moving Average Trend

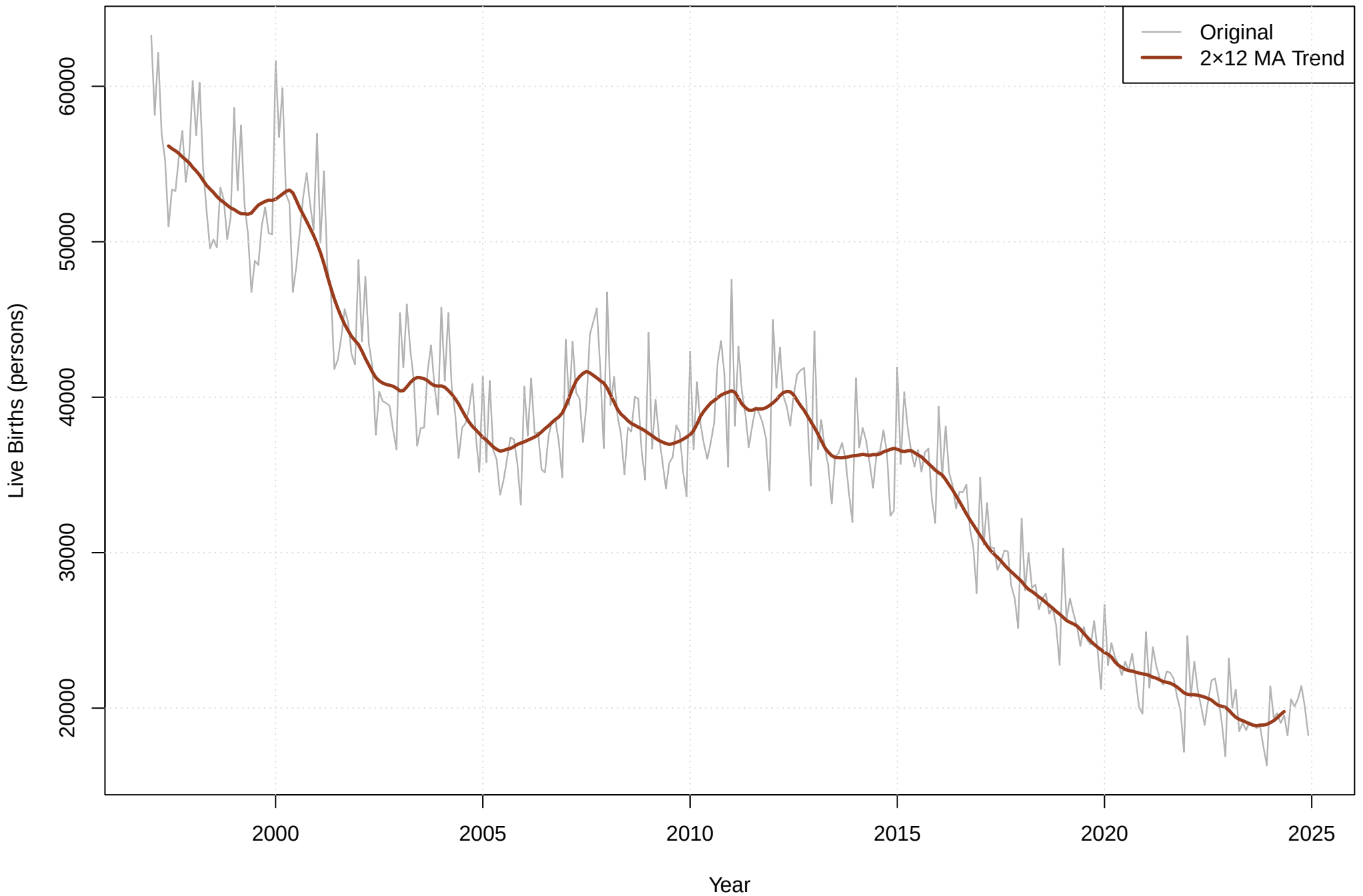


Fig 3. Detrended Series (Original $\div$ MA Trend)

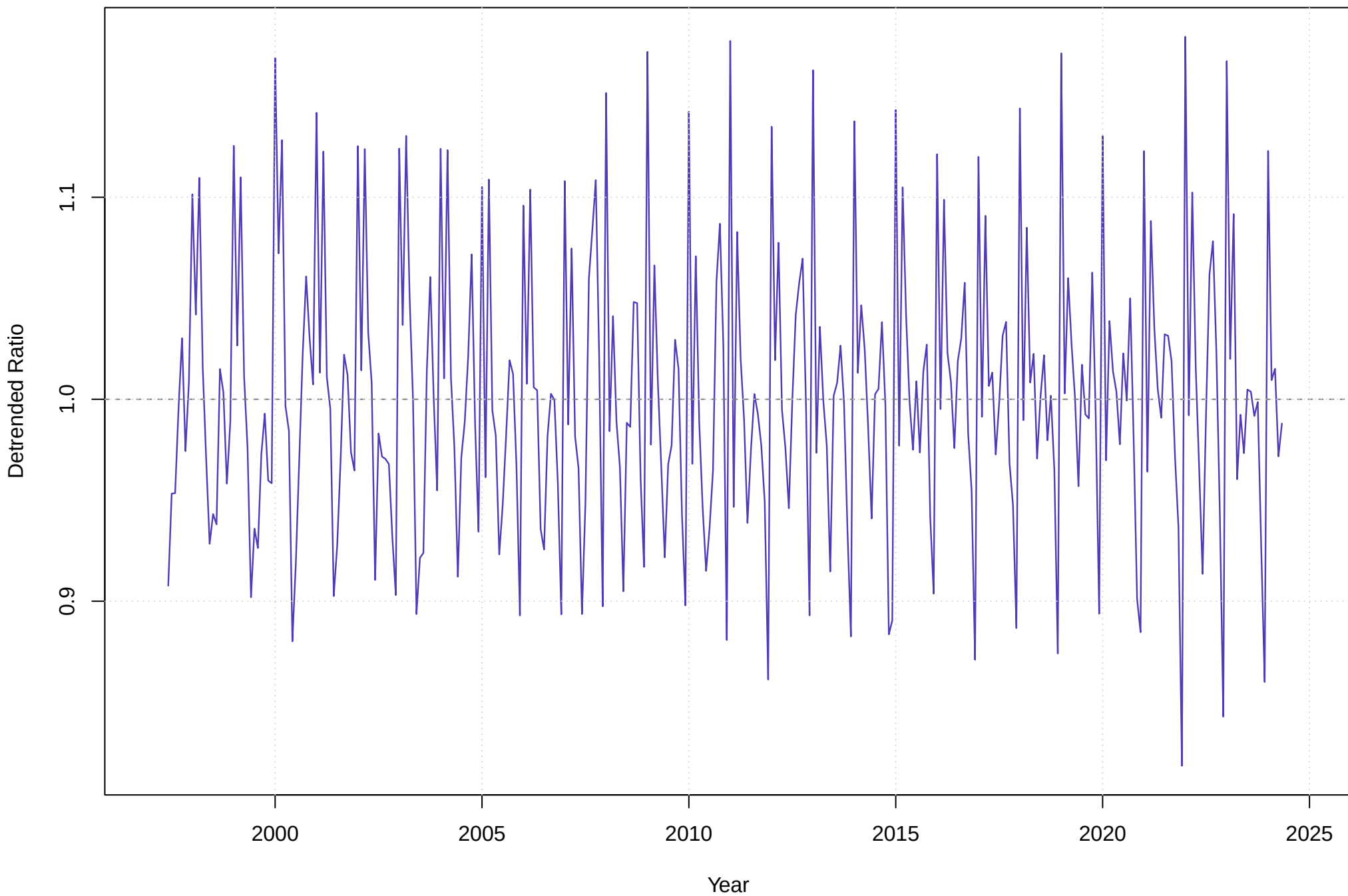


Fig 4. Monthly Seasonal Indices (Multiplicative)

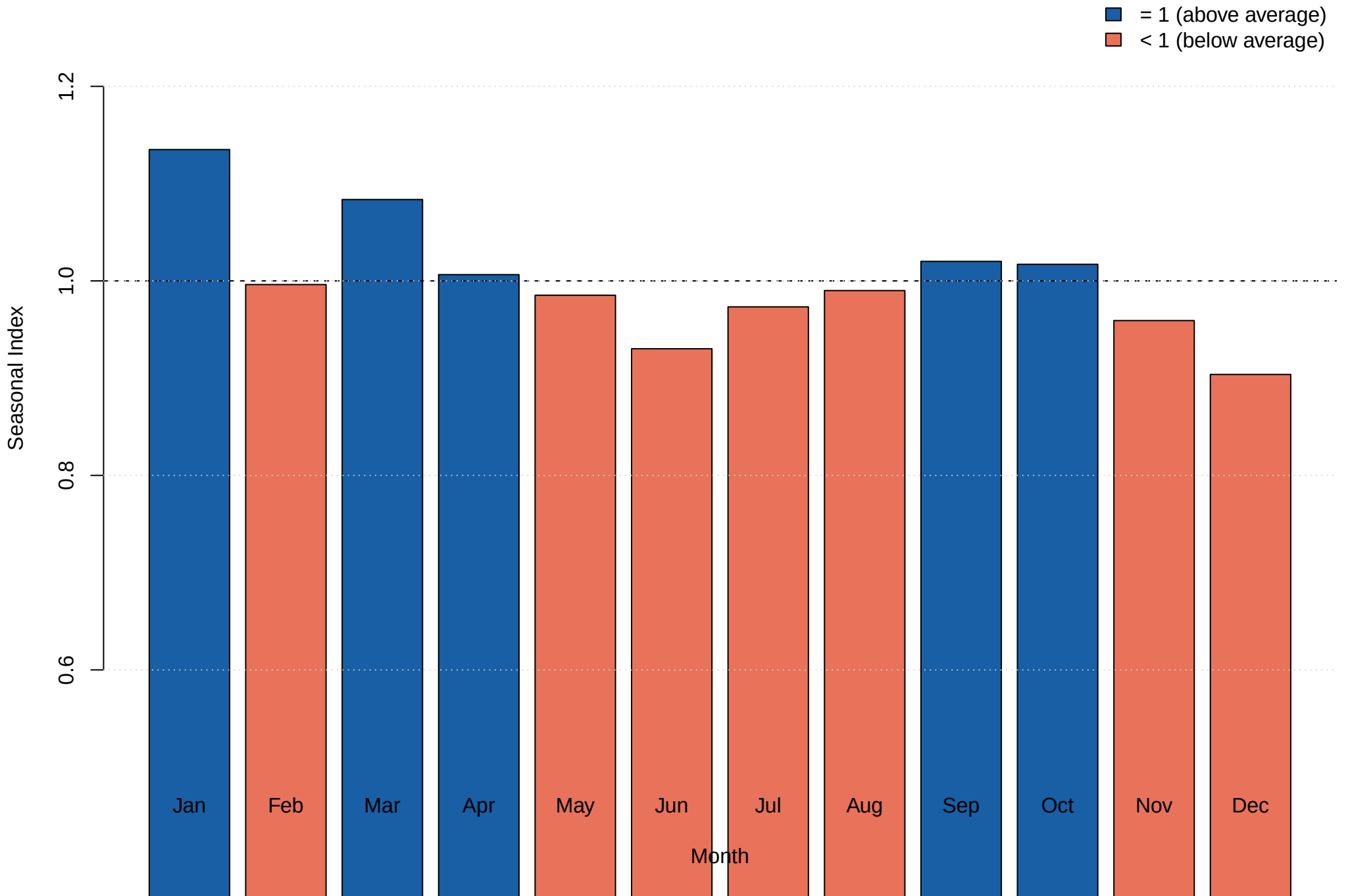


Fig 5. Deseasonalised Series (Trend + Remainder)

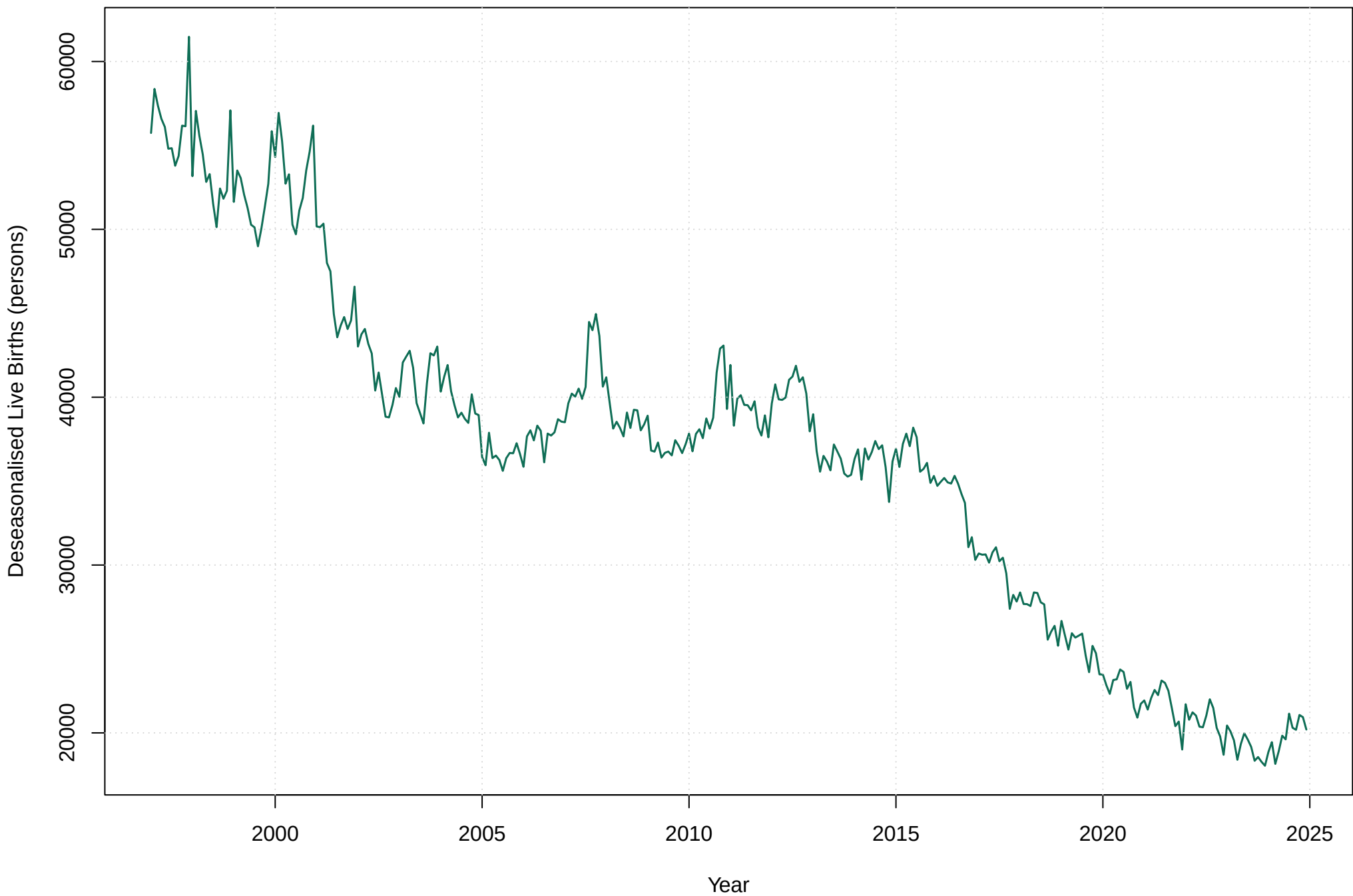
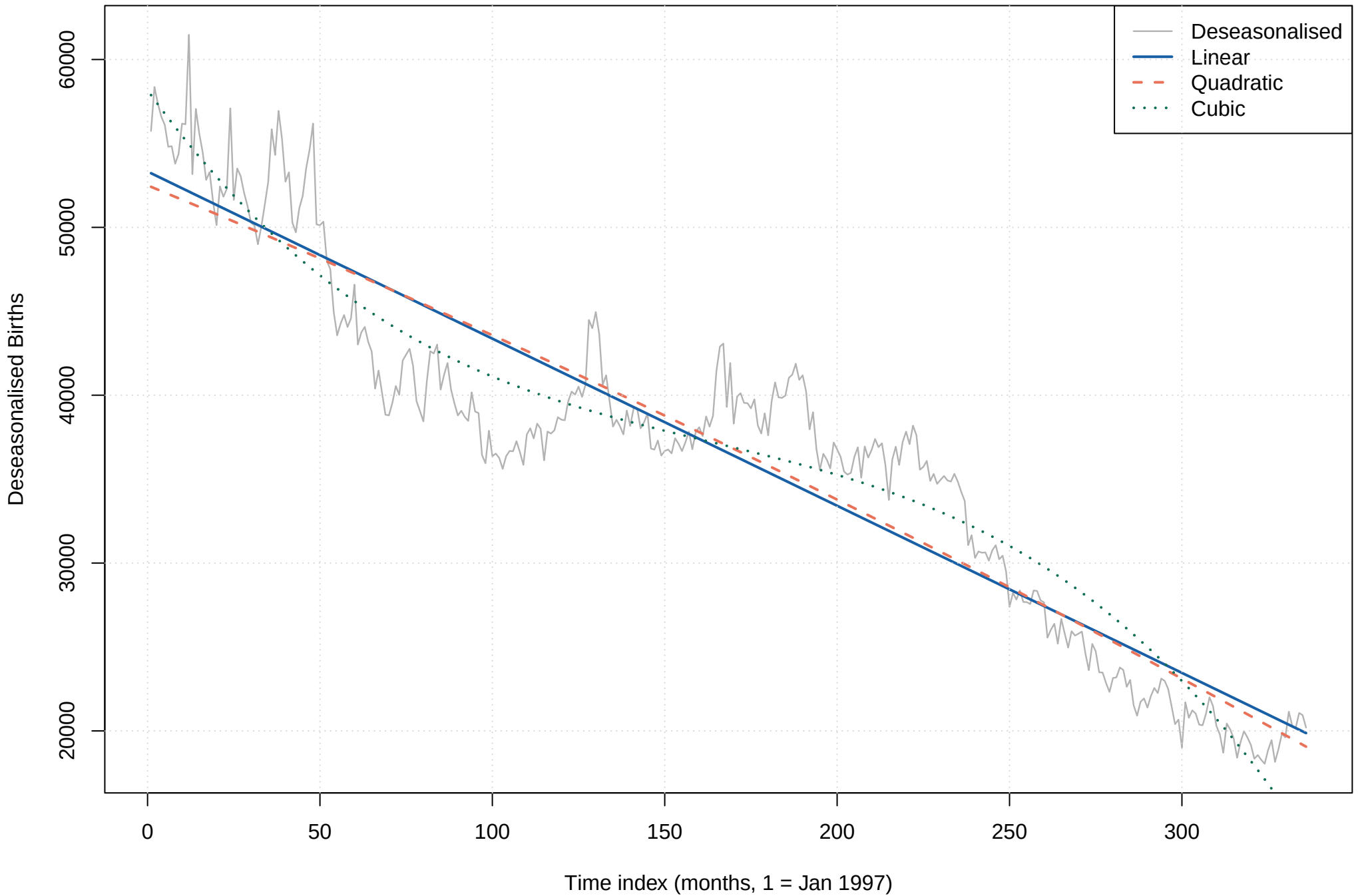


Fig 6. Deseasonalised Series with Fitted Trend Models



Decomposition of multiplicative time series

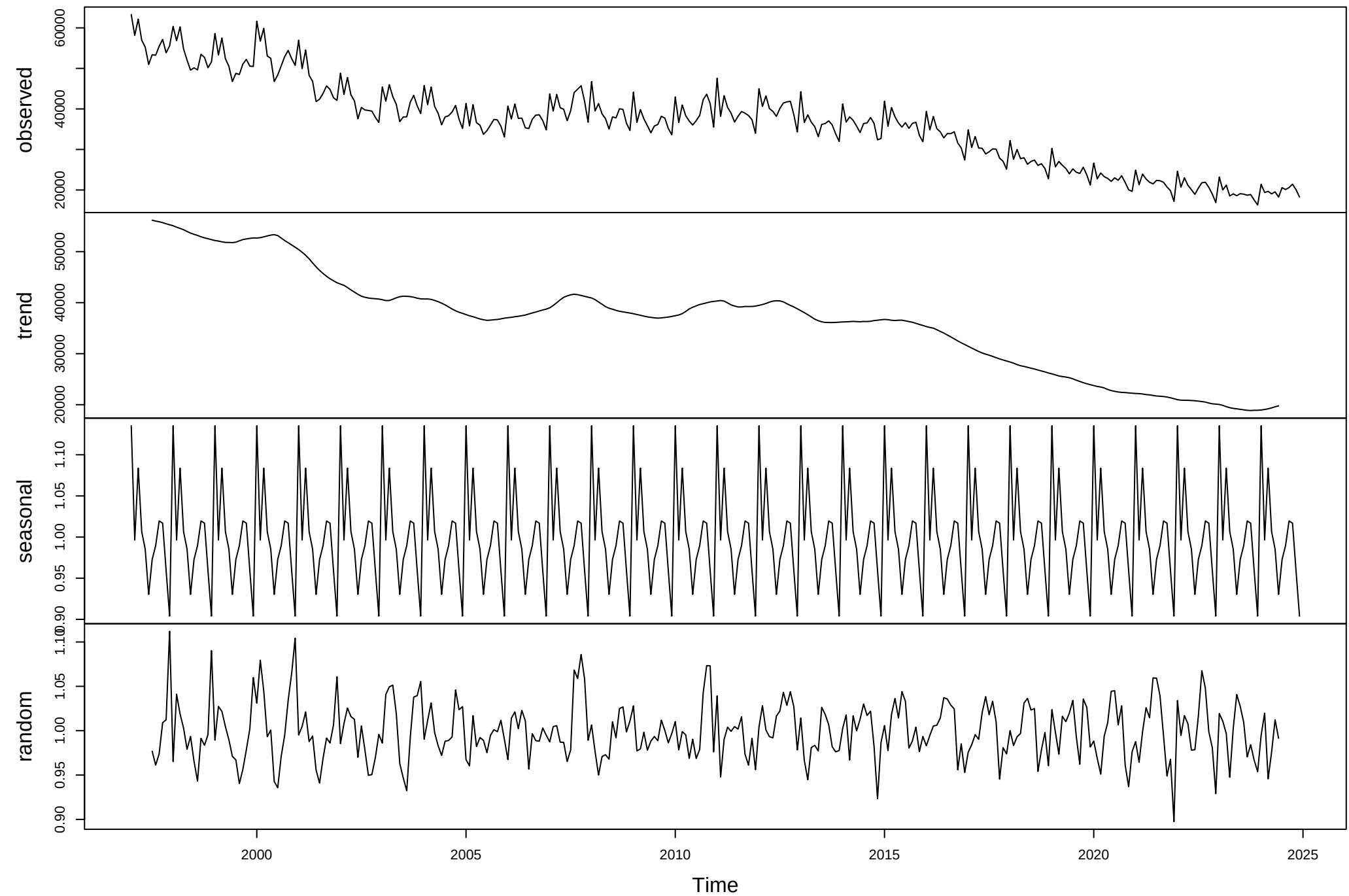


Fig 8. Stationary Series: Live Births after d=1, D=1 Differencing

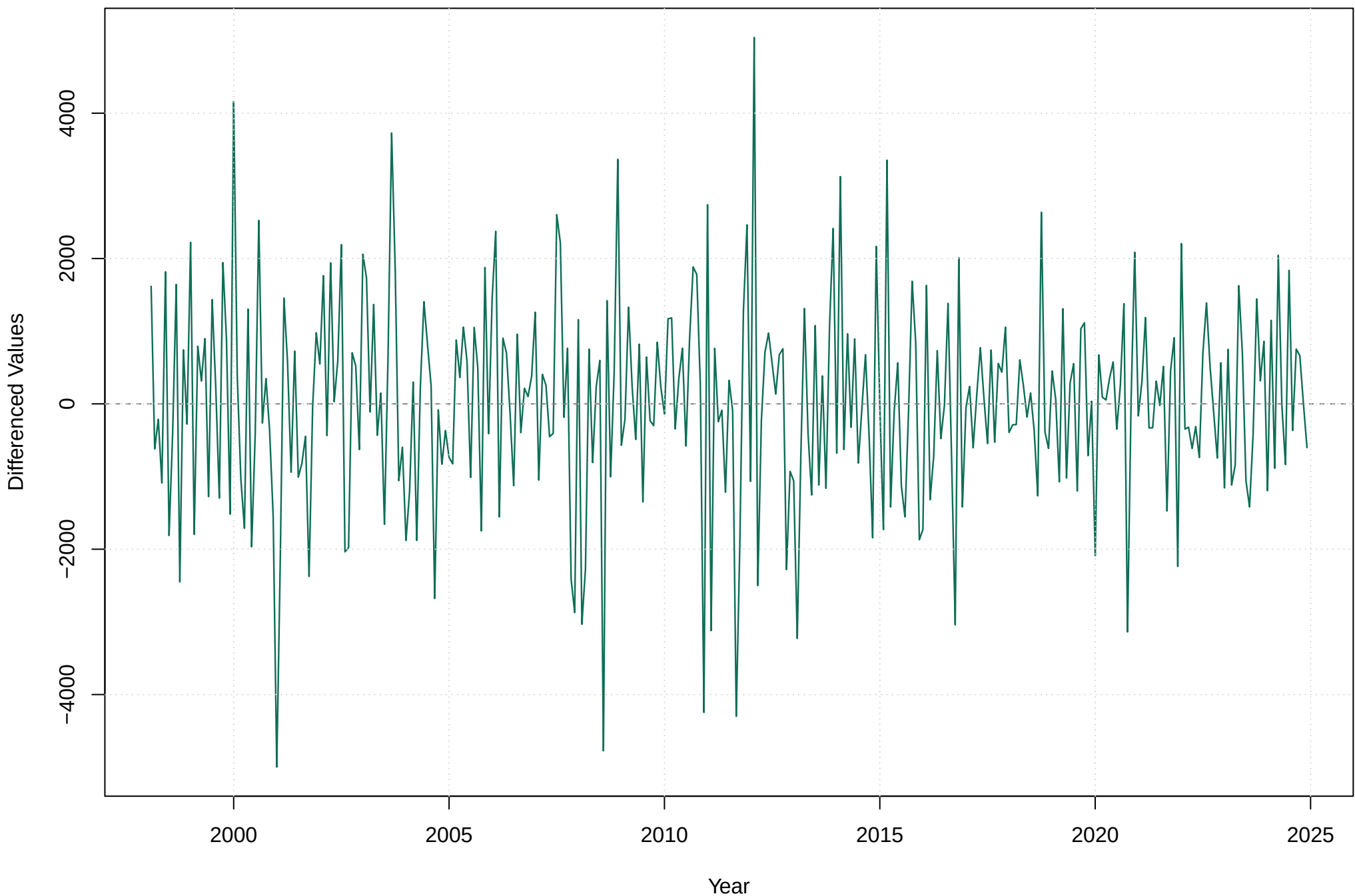


Fig 9a. ACF — Stationary Series (d=1, D=1)

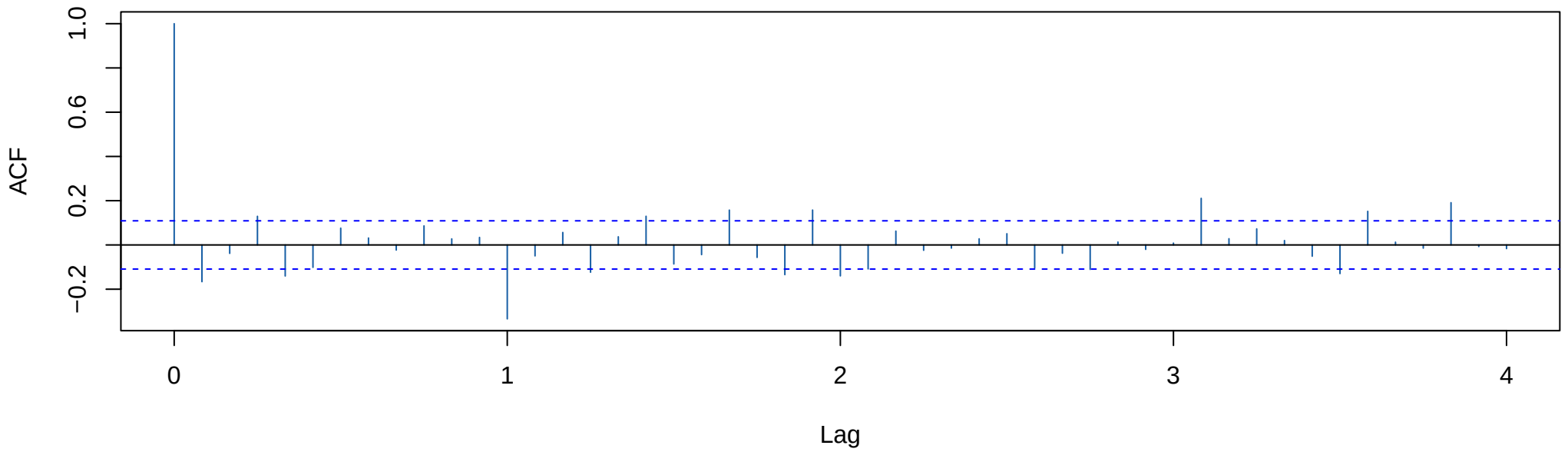
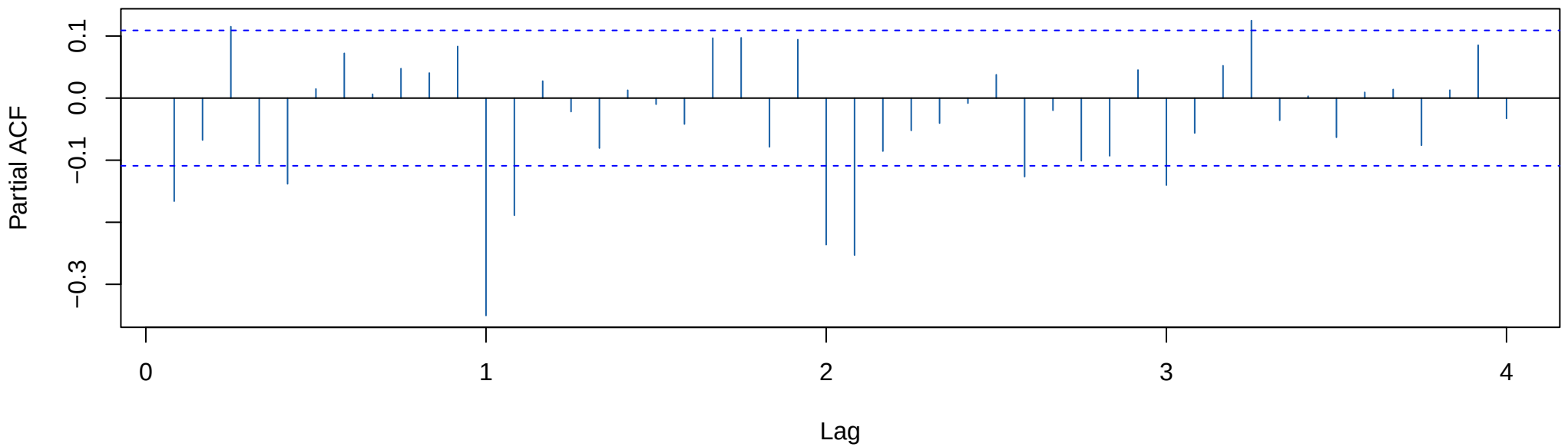


Fig 9b. PACF — Stationary Series (d=1, D=1)



Residuals from ARIMA(0,1,4)(0,1,1)[12]

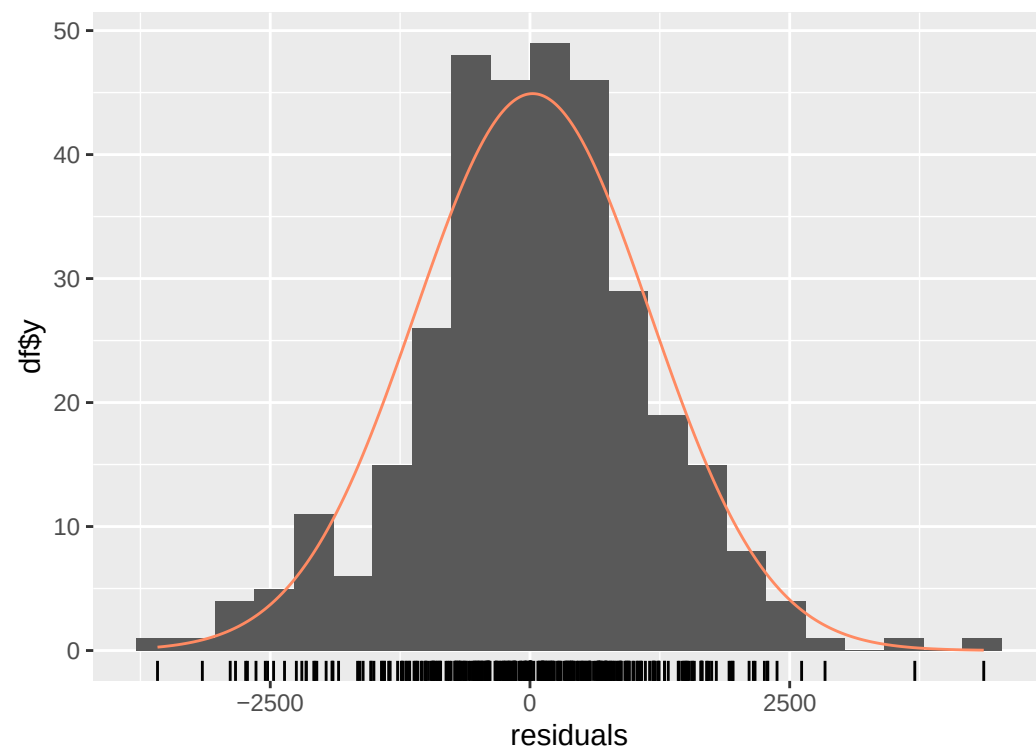
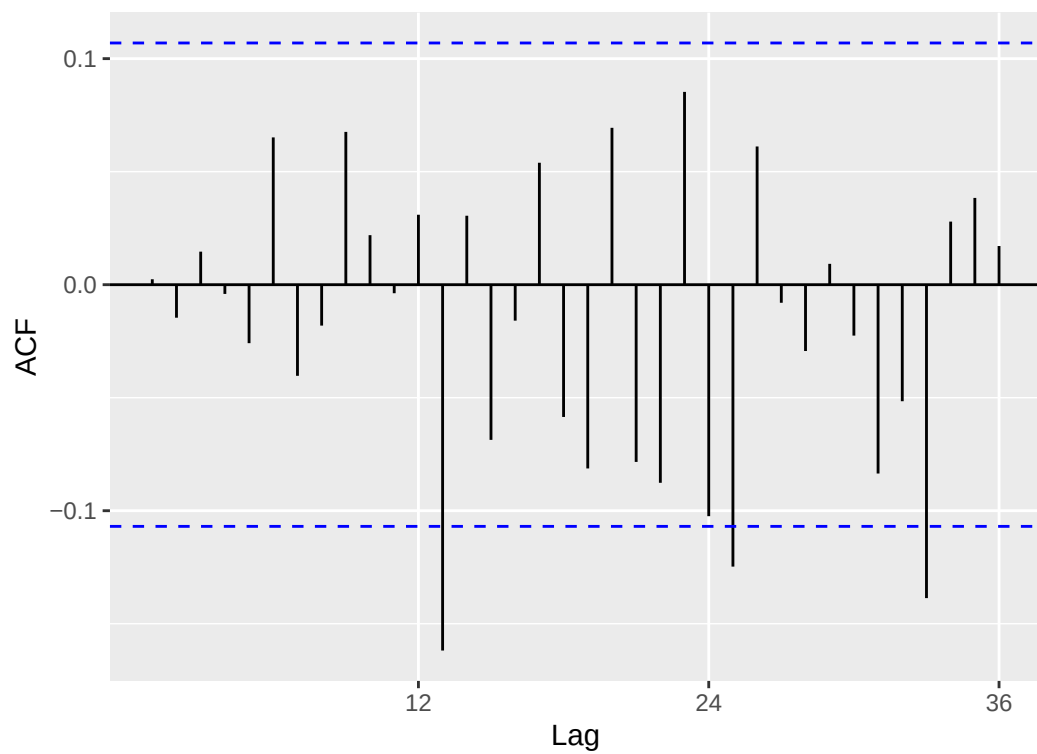
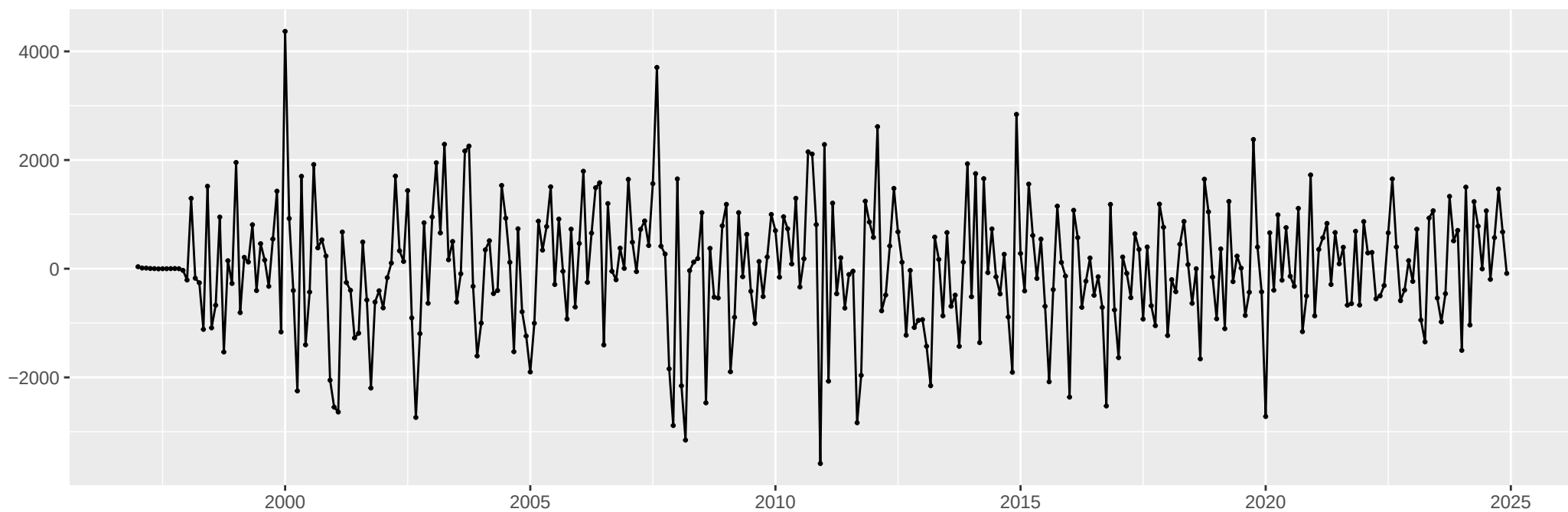
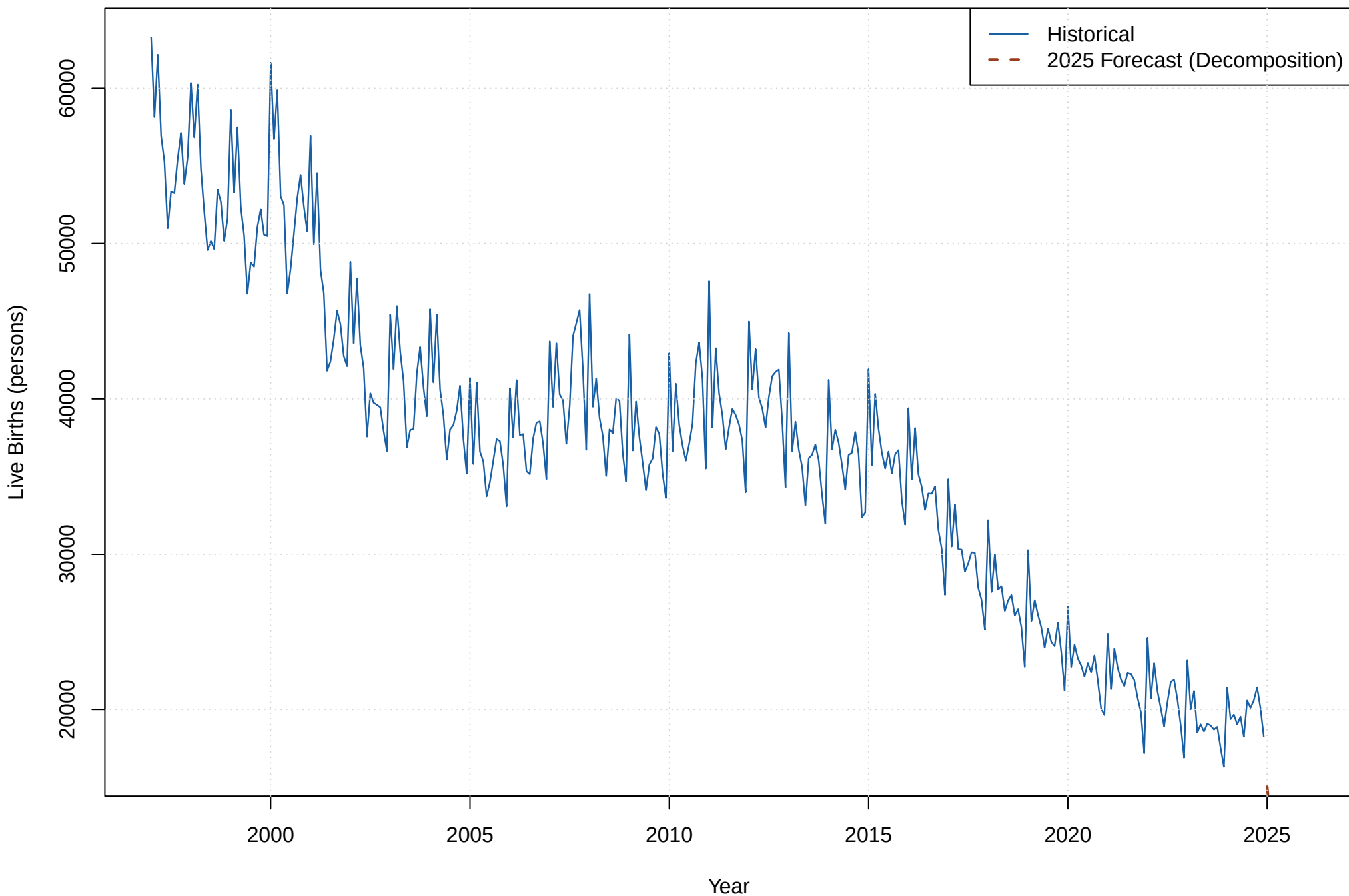


Fig 11. Live Births: Historical + 2025 Decomposition Forecast



5-Year Forecast of Live Births in South Korea (ARIMA, 2025–2029)

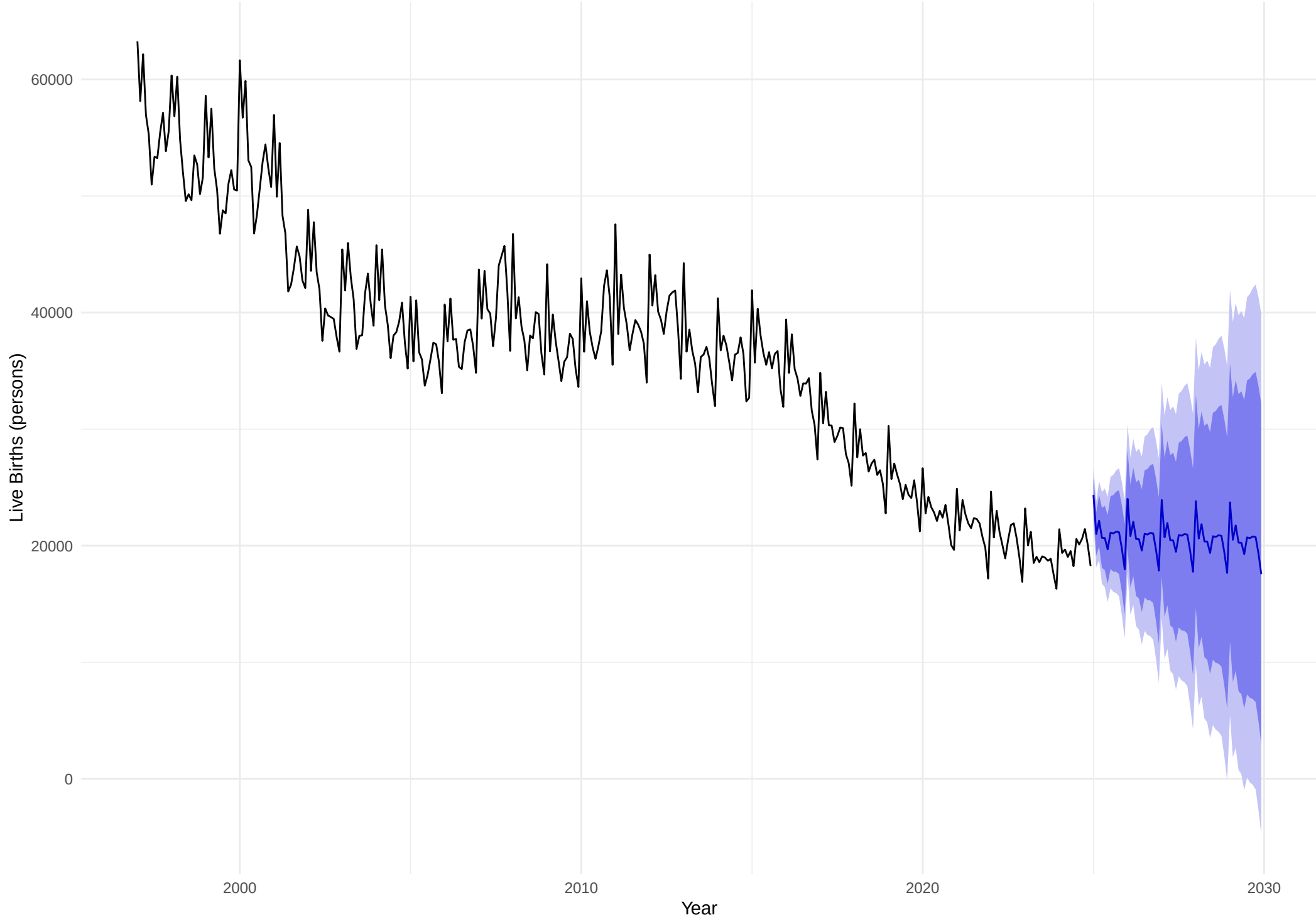


Fig 13. Historical + 2025 Forecasts: Decomposition vs ARIMA

